

Discrete Mathematics (MATH 208)

Fall 2022

Friday, 10/28/2022

Homework 9: Chapters 22, 23, 24

Score: / 9

Name (Print): _____

Chapter 22 - GCD's and the Euclidean Algorithm

2 pts

1. Use the Euclidean algorithm to compute $\gcd(a, b)$ in each case.

$$a = 216, \quad b = 111$$

$$a = 1001, \quad b = 11$$

$$a = 663, \quad b = 5168$$

$$a = 1357, \quad b = 2468$$

- 1 pts 2. If p is a prime, and n is any integer, what are the possible values of $\gcd(p, n)$?

- 1 pts 3. If p is a prime, and $0 < m \leq n$ are positive integers, determine $\gcd(p^m, p^n)$.

Chapter 23 - GCD's Reprised

- 1 pts 1. What is the smallest positive integer that can be written as a linear combination of 2191 and 1351?

2 pts

2. What are the possible values for $\gcd(a, b)$ if there exist integers s, t with $as + bt = 12$?

Chapter 24 - The Fundamental Theorem of Arithmetic

2 pts

1. Suppose n is an integer and that $n > 1$. Prove $2^{2n} - 1$ is not prime.